DATA FLOW DIAGRAM – EXPLANATION

1. Introduction

The Data Flow Diagram represents the movement of information between the user, local web application, authentication process and database in the authorized security lab.

2. Components

1. User: Sends login requests to the web application.
2. Web Application: Receives and processes user requests.
3. Authentication: Verifies user credentials.
4. Database: Stores and provides application data.
5. Trust Boundary: Separates the external user from the internal lab application components.

3. Data Flow

The user sends a login request to the local web application. The application forwards the credentials to the authentication process. The authentication process communicates with the database to verify the credentials. The verification result is returned to the application, which provides a response to the user.

LAB-02 is maintained as a separate isolated virtual machine asset in the authorized lab inventory.

4. Security Importance

The diagram helps identify trust boundaries, data movement and possible attack surfaces. It supports the identification of threats such as spoofing, tampering and information disclosure.

5. Conclusion

The Data Flow Diagram provides a clear understanding of the communication flow within the authorized lab environment and supports the STRIDE threat modelling process.